

Padding & Greedy Alignment

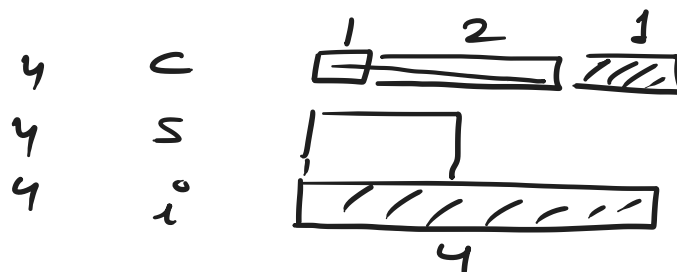
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Object Oriented Programming

```
class A {
    ...
}
```

padding

(24)



Greedy Algorithms:

1. Activity Selection Problem :

Given a set of activities with their start & finish times, select the maximum number of activities that can be performed by a single person, assuming that a person can only work on a single activity at a time.

1 4
A1,
5, 7
A4
8, 9
A5

Activity	Start Time	Finish Time
A1	1	4
A2	3	5
A3	0	6
A4	5	7
A5	8	9
A6	5	9

Steps: ① Sort the activities according to their finish time.

② We start by selecting the first activity which finishes earliest. For each activity, if the start time is after or equal to the finish time of the last selected activity, we select it.

③ The final selected activities are those which can be completed without overlapping.

Job Sequencing Problem

Problem Statement: →

Given a set of jobs where each job has a deadline associated with it, find the sequence of the jobs that yields the maximum profit. (Each job takes one unit of time). And, a job can only be scheduled before its deadline.

Job ID	Deadline	Profit
1	4	20
2	1	10
3	1	40
4	1	30

Sort the Jobs According to Profit

Job ID	Deadline	Profit
✓ 3	1	40
✓ 4x	1	30
✓ 1	4	20
✓ 2x	1	10

Slots = { none, none, none, none }
(J3, 40) none none (J1, 20)

(60)

Job ID	Deadline	Profit
1	2	100
2	1	19
3	2	27
4	1	25
5	3	15

Job ID	Deadline	Profit
1	2	100
✓ 3	2	27
4	1	25
2	1	19
5	3	15

1
100
27
15
142

Slots { none, none, none, none, none }
Slots { 1, 3, 5 }
3, 1, 5 ✓
1, 3, 5

3. Minimum Number of Coins

{ 1, 2, 5, 10, 20, 50, 100 } (93)

93 50, 20, 20, 2, 1 = 5

Greedy?

sort (asc) → { 5 → 1 }

$$93 - 100 = -7$$

50

$$93 - 50 = 43$$

C++

$$50 - 50 = 0$$

$$a - a[i] \geq 0$$

①

DP → Longest Common Subsequence

2D - DP

* Policeman & Thief